

Juan Francisco Martin

Software Engineer

Teguise, Canary Islands, Spain | juanframaro@gmail.com | +34 683 59 38 96

juanfrilla.github.io | linkedin.com/in/jfmozaga | github.com/juanfrilla

youtube.com/@jfmozaga

Summary

Experienced Web Scraping & Backend Engineer specializing in anti-bot bypasses, browser-based reverse engineering, and complex JavaScript deobfuscation (VM-based protections, Babel AST). I augment my development process with advanced, AI-assisted workflows (LLMs, MCPs).

Driven to step out of my comfort zone and eager for my first international role.

Education

Industrial Organization School (EOI), Bootcamp in Full Stack Web Programming – Madrid, Spain (Remote) Sept 2021 – Dec 2021

University of Las Palmas de Gran Canaria (ULPGC), BS in Telecommunications Engineering – Gran Canaria, Spain Sept 2012 – July 2019

- GPA: 7/10
- Technologies: Java, Matlab, C, Assembly, MySQL
- Final Project: Atmospheric sound classification system using ML and Direction of Arrival.

Experience

Web Scraping Engineer, Vumi.io – Asturias, Spain (Remote) Nov 2024 – Nov 2025

- Integrated and maintained a system with ~50+ scrapers (~900+ daily requests) to extract financial data from banks and brokers, growing a database of ~5 GB+.
- Reverse-engineered complex JavaScript using DevTools to automate MFA/2FA, JWT, and OAuth2 authentication flows via pure HTTP requests, avoiding browser overhead.
- Configured Selenium Grid and Scrapoxy (proxy network), on AWS EC2 with Docker, and Github Actions via SSH.
- Conected Playwright with Selenium Grid to support browser-based distributed and parallelized scraping tasks.
- I used AWS, that is, Lambda with Serverless architecture for scalable scraping, CloudWatch for observability, and S3 for CSV and JSON storage.
- Implemented advanced bypasses at scale for hCaptcha, reCAPTCHA v3 Enterprise, Cloudflare, Akamai, and Imperva.
- Extracted information from the DOM, APIs, or PDFs.
- Technologies: Burpsuite, Devtools, Scrapoxy, Selenium Grid, Linux, Docker, Git, AWS, PostgreSQL, Github Actions, Playwright, Pandas, Redis, Pytest

Web Scraping Engineer, Tirant Lo Blanch – Valencia, Spain (Remote) Feb 2023 – Nov 2024

- Integrated and maintained a system of ~700+ spiders (~70K+ daily requests) extracting legal data from government websites across Spain, Portugal, and LATAM, growing a database of ~10+ TB.
- Maintained a legacy project, written in the grails framework (groovy).
- Configured Selenium Grid on AWS EC2 to scale the browser infrastructure.
- Integrated a Tor Service proxy network for anonymized web scraping.
- Engineered a rotating proxies/headers system to avoid bans.
- Extracted data from the DOM, APIs, and created a system for parsing information from PDF, DOC, and DOCX files.
- Used AWS S3 for PDF storage.
- Implemented the scraping project's CI/CD with Gitlab CI via SSH.
- Technologies: Scrapy, Selenium Grid, Tomcat, Java, Groovy, Grails, MongoDB, SQL, Git, Linux, Docker, Gitlab CI, AWS

Back-end Developer, TuCalendi – Tenerife, Spain (Remote) Mar 2022 – Dec 2022

- Migrated and deployed a legacy PHP (CodeIgniter) API to Python (FastAPI) in Docker containerized environments, handling ~50+ concurrent calls with high availability.
- Designed and developed conversational voice applications for Amazon Alexa.
- Technologies: FastAPI, MariaDB, Linux, Git, Plesk, PHP, Docker, Github Actions, Postman, Pytest

Computer Technisian, Tías Town Hall – Lanzarote, Spain Mar 2021 – Sept 2021

- Technical support to employees and automation of cadastral tasks.

Languages

Spanish: Native

English: Professional Working Proficiency (C1)